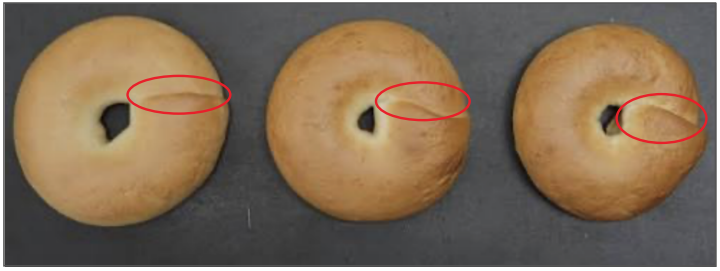
**Example industrial inputs; red = defect region**



bagel · crack



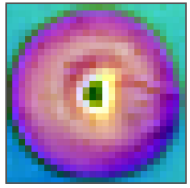
bolt · damaged thread (simulated)

RGB (bagel, crack)



DINO ViT-B/8 (frozen)

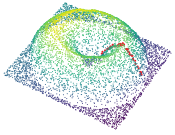
$E_{2D}$ 768×28×28 (PCA)



Crossmodal MLP mappers
$M_{2D\to 3D}$, $M_{3D\to 2D}$
(cosine loss, nominal data)

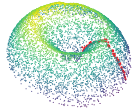
Inconsistency detector
$\Psi = \|E_{2D} - \hat{E}_{2D}\| \cdot \|E_{3D} - \hat{E}_{3D}\|$
anomaly map, $s = \max \Psi$

Organised point cloud (raw)

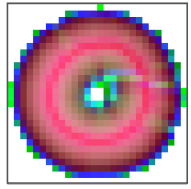


Preprocessing
A background removal
B RANSAC threshold
C XYZ interpolation

Cleaned point cloud (B = 5 mm)



Point-MAE (frozen)
D PFA neighbours k
E FPS centres G
F points/group S

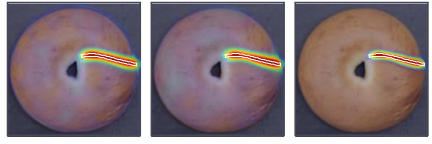


$E_{3D}$
1152×28×28
(PCA)

*optimum setting* ↑ *controllable factors (Taguchi L18)*

*illustrative maps*

$\Psi_{2D}$ $\Psi_{3D}$ $\Psi = \Psi_{2D} \cdot \Psi_{3D}$



Responses per run
I-AUROC,
AUPRO@30% (↑)
latency [ms] (↓)

Taguchi–Grey Relational Analysis
S/N ratios (10 categories) → main effects, ANOVA
grey relational grade → optimum levels A–F
prediction ± CI → confirmation runs